

Education

2022–Present **B.S. in Mathematics (Minor in Applied Mathematics)**, *University of Washington*, Seattle, WA, Dean's List (2023 Spr&Aut, 2024 Aut, 2025 Win&Spr&Aut, 2024–2025 Annual Dean's List)
Coursework: Two-year Honors Math Sequence (Vector Calculus, Differential Equations, Real & Complex Analysis), Abstract Algebra, Topology & Geometry, Probability, Database Systems, Artificial Intelligence, Machine Learning, and related topics.

Awards

Dec 2025 **Mary Gates Research Scholarship (\$5000)**
Oct 2025 **ICDM Student Travel Award (Funded by NSF)**
Mar 2026 **University of Washington Undergraduate Research Conference Travel Award**

Internship Experience

- Jun 2025 – **Prompt Engineering Intern, MaaS Department, Zhipu AI**, Beijing, China
- Aug 2025
- Contributed to the **REDNOTE** (a social media platform) **AI translation system**, aiming for fluent and culturally appropriate cross-lingual communication; focused on **multilingual prompt design**, **RAG pipelines**, evaluation, and translation strategy/pipeline design, with extensive hands-on work in **Dify**.
 - Designed and deployed multiple **Dify workflows** inspired by top-conference papers, including evaluation suites for in-house models and **instruction fine-tuning** of multimodal LLMs (GLM-4v/4.5v); supported B2B use cases such as extracting structured outputs from PDF/image contracts.
 - Authored the **PRD** and built the **MVP** for an LLM prompt-sharing website; developed workflows for prompt-to-HTML rendering, multi-speaker podcast generation, and TTS stress testing.
- May 2024 – **Data Science Intern, RealAI**, Beijing, China
- Aug 2024
- Worked on **“Jingce”**, an AI policy-advising platform for government policy drafting and revision (compliance checks, historical policy search, policy suggestions, consistency checks, and rewriting) built on domain policy corpora.
 - Under the supervision of a senior data engineer, built large-scale pipelines to structure unstructured web data: automated crawling, cleaning, anomaly detection/handling, normalization, missing-value processing, and cross-table validation using **Python** (*pandas*, *NumPy*) and **SQL**; audited dataset conflicts and errors.
 - Processed a dataset of roughly **100K** semi-structured records with about **50 attributes** per record, emphasizing reliability and automated quality control for model inputs.
 - Conducted prompt-engineering experiments based on published papers to improve **context retention** and multi-turn dialog quality for long-horizon human–AI interaction during policy drafting.
- Aug 2024 – **Cybersecurity Intern, Security BU, Qi An Xin (QAX)**, Beijing, China
- Oct 2024
- Participated in a large-scale red-blue cybersecurity exercise as part of a rotating blue-team monitoring/response unit (12/7 shift), defending the core assets of a large central state-owned mining group against expert red teams.
 - Monitored network/server traffic, triaged alerts (true positives vs. false positives), reported incidents, and supported operational response procedures.
 - Outside the exercise, used **pandas** to assist HR with operational data processing for QAX cybersecurity services (2023 data).

Research Experience

Nov 2024 – **Undergraduate Researcher**

Present **High-Performance Data-Intelligence Computing (HPDIC) Lab**, *University of Washington*, Seattle, WA, Advisor: Prof. Dongfang Zhao

- **ProHD: Projection-Based Hausdorff Distance Approximation (ICDM 2025)** — proposed a projection-based subset-selection approach for fast Hausdorff distance estimation in high dimensions, using centroid directions and principal components to guide selection; achieved 10–100× **speedups** with controllable error on multiple datasets; supported by **PNNL** and evaluated on the Texas A&M ACES cluster and Chameleon testbed; presented in Washington, D.C.
- **QPAD: Quantile-Preserving Approximate Dimension Reduction for Nearest Neighbors Preservation in High-Dimensional Vector Search (ICDE 2026)** — identified recall degradation in ANN search after applying common DR methods (e.g., PCA/UMAP) and developed an **unsupervised** dimensionality-reduction method that improves neighborhood preservation by enlarging distances between nearby vectors; provided rigorous proofs of key mathematical properties and evaluated against baselines on **six datasets**, showing consistent accuracy gains for mainstream ANN methods.
- **Directed Hypergraph Compression and Reordering** — studying a 1D array storage format for **directed hypergraphs** to improve compression, cache locality, and runtime of hypergraph algorithms in ML/database workloads.
- **Techniques and Stack: Linux/VM; C/C++ (STL, OpenMP, Makefile); Python (multiprocessing, psutil, pandas, NumPy, SciPy); FAISS (ANN search); matplotlib; mpi4py and PyOpenCL; Conda.**

Mar 2025 – **Paid Research Assistant**, *School of Vehicle and Mobility, Tsinghua University*, Beijing, China

- Present
- **SSIDN / CI-CNN (i2MTC 2026)** — generated high-quality training segments via **fault injection** on real-world time-series data from railway turnout actuators and industrial-robot sensors to improve robustness to sensor failures.
 - Simulated sensor-abnormality patterns including **short-circuit**, **open-circuit**, **periodic interference**, **impulse noise**, and **sensor tilt**; supported ablation studies and feature visualization.
 - Built the end-to-end error-injection and time-series processing pipeline in **Python** (*NumPy, pandas, PyTorch*).

Dec 2023 – **Undergraduate Researcher**

Dec 2024 **Washington Experimental Mathematics Lab (WXML)**, *University of Washington*, Seattle, WA

- Studied the **Concentration of Measure** phenomenon in measure theory/probability/combinatorics; derived analytic asymptotics and probabilistic limits via the **Curie–Weiss model**, and validated results using **Monte Carlo simulation** and symbolic computation in *Wolfram Mathematica*.
- Presented a poster at the **UW Undergraduate Research Symposium**. [Poster link](#)

Publications

- ICDM 2025, accepted on 8/25/2025. **Jiuzhou Fu**, Luanzheng Guo, Nathan Tallent, Dongfang Zhao. *ProHD: Projection-Based Hausdorff Distance Approximation*. **IEEE International Conference on Data Mining (ICDM) 2025** (full research paper, full-length oral presentation, acceptance rate: 13%). <https://arxiv.org/abs/2511.18207>
- ICDE 2026, accepted on 2/23/2026. **Jiuzhou Fu**, Dongfang Zhao. *QPAD: Quantile-Preserving Approximate Dimension Reduction for Nearest Neighbors Preservation in High-Dimensional Vector Search*. **IEEE International Conference on Data Engineering (ICDE) 2026**. <https://arxiv.org/abs/2504.16335>
- i2MTC 2026, accepted on 2/25/2026. Yuhan Huang, Xiaoxi Hu, **Jiuzhou Fu**, Fei Chen, Jingming Cao, Naibang Wang. *SSIDN: A Small-Sample Incremental Diagnosis Network for Railway Turnout Actuators*. **IEEE Instrumentation and Measurement Technology Conference (i2MTC) 2026** (Flagship conference of IEEE Instrumentation and Measurement Technology Society). [paper](#)
- i2MTC 2026, accepted on 2/25/2026. Xu Zhang, Xiaoxi Hu, Yuhan Huang, **Jiuzhou Fu**, Jingming Cao, Tao Tang. *Channel-Incremental Convolutional Neural Network for Railway Point Machine Fault Diagnosis with Expanding Sensor Configurations*. **i2MTC 2026**. [paper](#)